Ex. No. : 03 **Date:**

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A PYTHON PROGRAM TO IMPLEMENT LOGISTIC MODEL

Aim:

To implement a **Logistic Regression model in Python** to classify **Iris dataset**

Requirements:

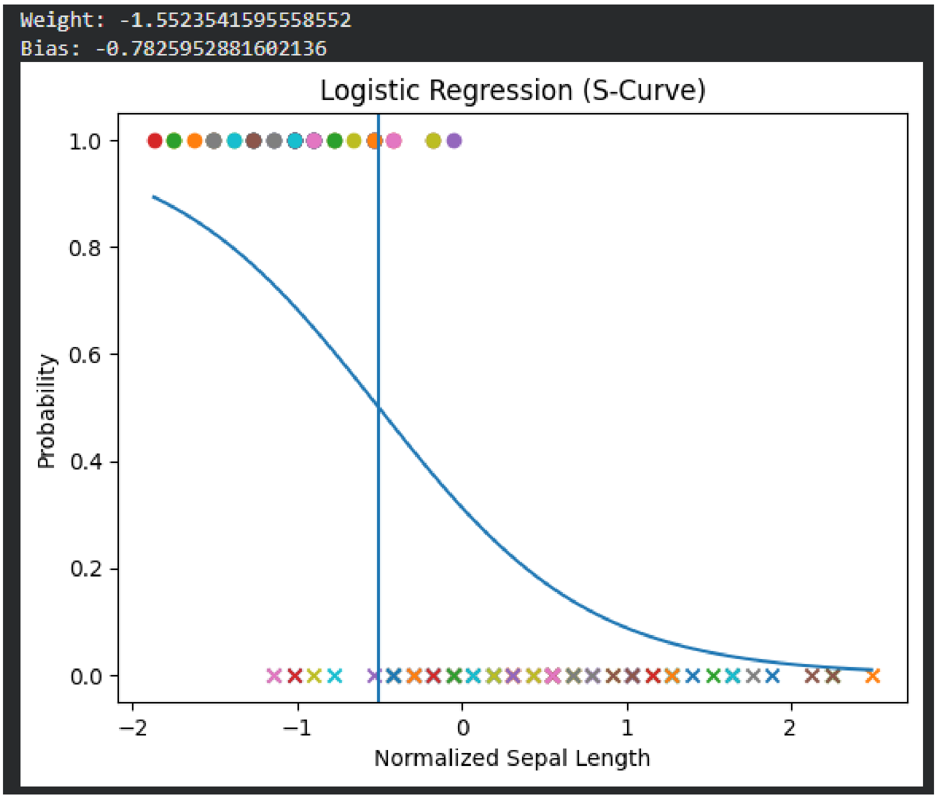
Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()

df = pd.read_csv('Iris.csv')

df['target'] = df['Species'].apply(lambda x: 1 if x == 'Iris-setosa' else 0)

X = df['SepalLengthCm'].values
Y = df['target'].values

X = (X - X.mean()) / X.std()

w = 0
b = 0
lr = 0.01
epochs = 1000

def sigmoid(z):
   return 1 / (1 + np.exp(-z))
```

```
for _ in range(epochs):
    z = w * X + b
    y_pred = sigmoid(z)

    dw = np.mean((y_pred - Y) * X)
    db = np.mean(y_pred - Y)

    w -= lr * dw
    b -= lr * db

print("Weight:", w)
print("Bias:", b)

x_line = np.linspace(min(X), max(X), 200)
y_line = sigmoid(w * x_line + b)

plt.figure()

for i in range(len(X)):
    if Y[i] == 1:
        plt.scatter(X[i], Y[i])
    else:
        plt.scatter(X[i], Y[i], marker='x')

plt.plot(x_line, y_line)

decision_boundary = -b / w
plt.axvline(x=decision_boundary)

plt.title("Logistic Regression (S-Curve)")
plt.xlabel("Normalized Sepal Length")
plt.ylabel("Probability")

plt.show()
```

OUTPUT:



Result:

Thus, the **Logistic Regression model was successfully implemented using Python and evaluated using the Iris dataset.**